

04 - DOM Extraction Engine Spec

Visor DOM-to-Agent Context Compiler
DOC-04 | 0.1 requirements pack | Generated 2026-06-14

This specification defines how the content script extracts a safe intermediate PageSnapshot from a live webpage DOM.

Field	Value
Owner	Rohan PX / Visor project
Audience	AI coding agent, reviewer, future contributors
Status	Draft baseline for MVP build
Decision rule	MVP is local-first, minimal-permission, Chrome Manifest V3, no backend unless explicitly enabled.

Document control

Item	Value
Project	Visor DOM-to-Agent Context Compiler
Document code	DOC-04
Generated on	2026-06-14
Primary goal	Turn raw webpage DOM into safe, structured, agent-ready context.
MVP boundary	Chrome extension, local-only compiler, JSON/Markdown export, privacy redaction, tests.

1. Extraction goal

The extractor creates an intermediate PageSnapshot from the active page. It must preserve content needed for an AI agent while removing or labeling material that is hidden, duplicated, irrelevant, unsafe, or non-semantic.

2. Pipeline

Stage	Input	Output	Notes
1. Snapshot	Live DOM	Candidate nodes	Traverse document body; avoid script execution.
2. Visibility filter	Candidate nodes	Visible nodes	Use computed style, bounding box, aria-hidden, hidden attribute.
3. Semantic extraction	Visible nodes	Blocks	Headings, text, links, tables, forms, buttons, media, code.
4. Context attachment	Blocks	Blocks with surroundings	Attach nearby heading, section, form label, table caption.
5. Noise marking	Blocks	Kept/removed candidates	Mark rather than delete in debug mode.
6. Snapshot validation	PageSnapshot	Valid snapshot or error	Use schema validation before compiler.

3. Preserve list

- Document title, URL, canonical URL, meta description, OpenGraph title/description where useful.
- H1-H6 headings with hierarchy level and DOM order.
- Paragraphs, list items, blockquotes, article text, visible labels, error messages, status messages.
- Links with visible text, href, title, rel, and nearby context.
- Buttons and role=button elements with aria-label/title/text/nearby context.
- Forms, inputs, selects, textareas, labels, placeholders, required flags, disabled state, validation text.
- Tables including headers, captions, row/column values, and surrounding heading.
- Code blocks, pre tags, inline code in documentation contexts.
- Images only as metadata: alt text, caption, src domain, dimensions if relevant.

4. Remove or down-rank list

- script, style, noscript, template, tracking pixels, invisible SVG noise.
- Hidden elements: display none, visibility hidden, opacity zero with no interaction role, hidden attribute, aria-hidden true.
- Cookie banners, ad containers, newsletter popups, social share widgets, generic footer legal links.

- Repeated navigation, duplicated breadcrumbs, repeated card templates, sidebars with unrelated content.
- Empty elements and text shorter than a configurable threshold unless action labels require them.

5. Visibility heuristic

```
function isProbablyVisible(el: Element): boolean {
  const style = window.getComputedStyle(el);
  const rect = el.getBoundingClientRect();
  if (style.display === 'none' || style.visibility === 'hidden') return false;
  if (el.hasAttribute('hidden') || el.getAttribute('aria-hidden') === 'true') return false;
  if (rect.width === 0 && rect.height === 0 && !isSemanticInput(el)) return false;
  return true;
}
```

6. PageSnapshot schema

```
type PageSnapshot = {
  schemaVersion: 'page_snapshot.v1';
  source: {
    url: string;
    canonicalUrl?: string;
    title: string;
    capturedAt: string;
    language?: string;
  };
  metadata: Record<string, string>;
  headings: HeadingBlock[];
  textBlocks: TextBlock[];
  links: LinkBlock[];
  actions: ActionBlock[];
  forms: FormBlock[];
  tables: TableBlock[];
  media: MediaBlock[];
  codeBlocks: CodeBlock[];
  stats: DOMStats;
  warnings: ExtractionWarning[];
};
```

7. Edge cases

Case	Expected behavior
Shadow DOM	Traverse open shadow roots when safe; report closed shadow roots as inaccessible.
Iframes	Same-origin iframe may be inspected; cross-origin iframe must be reported as inaccessible.
Infinite scroll	MVP extracts currently loaded DOM only; do not auto-scroll.
Dynamic React apps	Extract visible rendered DOM; do not infer hidden application state.
Very large DOM	Apply node cap, block cap, and partial snapshot warning.
Canvas-only page	Return metadata and warning that semantic DOM content is limited.

Source and reference notes

These sources are used to ground platform/security assumptions. The implementation must still verify current platform behavior during development.

Reference	Why it matters
MDN WebExtensions - Content scripts: https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions/Content_scripts	Grounding for content script behavior, host permissions, and page DOM access.